

Andrea Dunn Beltran

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Education

University of North Carolina at Chapel Hill

Chapel Hill, NC

Master of Science in Computer Science

Spring 2024 – Present

Relevant coursework: AI; 3D Generative Models; Visual Recognition with Transformers; Computational Biology; Mobile Health Systems; Deep Learning; Robot Learning

University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Computer Science & Applied Mathematics

Fall 2020 – Fall 2023

Relevant coursework: Algorithms; Machine Learning; Neural Rendering; Probabilistic Models & Deep Structured Prediction; 3D Computer Vision; Optimization for ML; Probability

Publications and Manuscripts

- **Andrea Dunn Beltran***, Daniel Rho*, Marc Niethammer, Roni Sengupta. [NFL-BA: Near-Field Light Bundle Adjustment for SLAM in Dynamic Lighting](#). *NeurIPS 2025*.
- **Andrea Dunn Beltran***, Romain Hardy*, Pranav Rajpurkar. [Seeing down the Line: FINISH NAMING PAPER](#). *Under Review 2025*.
- Romain Hardy, **Andrea Dunn Beltran**, Todd Brenner, Tyler Berzin, Pranav Rajpurkar. [Rendering With a Gut Feeling: Depth-Guided Splatting for Complete Colonoscopic Scene Reconstruction](#). *Under Review 2025*.
- Xinqi Xiong, **Andrea Dunn Beltran**, Jun Myeong Choi, Marc Niethammer, Roni Sengupta. [PPS-Ctrl: Controllable Sim-to-Real Translation for Colonoscopy Depth Estimation](#). *CVPR MMFM-BIOMED Workshop*.
- Noah Frahm, Dongxu Zhao*, **Andrea Dunn Beltran***, Ron Alterovitz, Jan-Michael Frahm, Junier Oliva, Roni Sengupta. [VIN-NBV: A View Introspection Network for Next-Best-View Selection for Resource-Efficient 3D Reconstruction](#). *arXiv 2025*.
- Frances Bashore, Ariana Marquez, Apirat Chaikuad, Stefanie Howell, **Andrea Dunn Beltran**, Álvaro Beltran, Jeffery Smith, David Drewry, Adriana Beltran, Alison Axtman. [Modulation of TTBK1 and TTBK2 Impacts Ciliogenesis](#). *Scientific Reports 2023*.

Research Experience

Visiting Graduate Student, Harvard Medical School

Fall 2023 – Present

Supervisor: Prof. Pranav Rajpurkar

Geometry-Aware Robust Endoscopic Reconstruction

- Leading a new SLAM formulation that incorporates colon centerline geometry into bundle adjustment via coupled camera-centerline and map-centerline consistency losses.
- Implemented spline-based anatomical priors and differentiable centerline regularization for improved trajectory consistency and global scene coherence.
- Preparing first-author MIDL submission describing the method and evaluation.

Research Assistant, UNC Chapel Hill

Spring 2023 – Present

Supervisor: Prof. Roni Sengupta

Next-gen 3D Modeling of Endoscopy Videos

- Lead developer for 4D bronchoscopy reconstruction pipeline; implemented synthetic breathing-lung dataset generation and deformable reconstruction framework.
- Designed the Near-Field Lighting Bundle Adjustment (NFL-BA) algorithm, integrating per-pixel lighting cues into dense SLAM and achieving state-of-the-art pose accuracy on C3VD.
- Contributed evaluation tooling to PPS-Ctrl (Stable Diffusion + ControlNet), enabling photorealistic sim-to-real colonoscopy translations and improved depth estimation.
- Supported optical-flow and reflectance-based exploratory algorithms; performed data curation and validation for early-stage methods.

Task-Aware 3D Knowledge Acquisition

- Co-author of VIN-NBV: implemented baseline networks and benchmarking suite; contributed to a next-best-view policy improving reconstruction quality by 30%.
- Designed data-preparation and evaluation pipelines; conducted literature synthesis for project framing.

Undergraduate Research Assistant, University of Southern California

Summer 2023

NSF REU: Robotics & Autonomous Systems — *Reinforcement Learning with Temporal Logic*

- Developed Python-based Gazebo and OpenAI Gym environments for RL training on the Toyota HSR platform.
- Investigated reinforcement learning under temporal-logic constraints; built data pipelines for imitation learning.
- Presented findings at the SoCal REU Symposium.

Undergraduate Research Assistant, Rice University

Summer 2022

REU: Data Science — *Online Learning for Medical Imaging*

- Built online-streaming pipeline and custom TensorFlow CNNs for class-imbalanced chest X-ray datasets.
- Conducted ablations on learning-rate schedulers and indexing to mitigate catastrophic forgetting.
- Automated augmentation and continual-learning workflows; presented results at the Rice REU Symposium.

Department of Pharmacology Research Intern, UNC Chapel Hill

Summer 2018 – Spring 2022

BRAC2 Mutations and Modulation of Tau Tubulin Kinases

- Performed immunofluorescence staining and high-resolution imaging of 200+ cilia per month.
- Built ImageJ macros and Python segmentation scripts, reducing analysis time by 80%.
- Executed dose-response and viability assays on TNBC cell lines; confirmed apoptosis via caspase-3 cleavage.
- Gained experience in sterile culture, Western blotting, plasmid prep, bacterial transformation, fluorescence microscopy.

Teaching and Mentoring

Graduate Teaching Assistant, Department of Computer Science

UNC Chapel Hill

COMP 590/776: *Computer Vision in the 3D World*

Fall 2023, 2024, 2025

COMP 116: *Introduction to Scientific Programming*

Spring 2024

Undergraduate Teaching Assistant, Department of Applied Science and Engineering

UNC Chapel Hill

APPL 101: *Introduction to Engineering*

Fall 2022

Mentorship: MIT Summer Geometry Initiative: SLAM (2024); UNC–Intel REU: Deep-Fake Detectors (2024); Graduate Women in CS (2024, 2025); Undergraduate Researchers (2025)

Skills and Achievements

Programming

Python; MATLAB; C; Java; JavaScript; SQL; R; MIPS; HTML

Machine Learning & Vision

PyTorch; TensorFlow; OpenCV; Open3D

Software & Tools

Linux/Unix; Git; Docker; CUDA; Blender; 3D Slicer; ImageJ; VTK/ITK

Robotics & Simulation

ROS; Gazebo; OpenAI Gym

Awards

CVPR BP/DEI Scholarship (2024, 2025); NSF REU Supplement (2023); Dean's List (2021–2023); Hayden B. Renwick Scholar (2021–2023)

Leadership, Outreach, and Service

Department Roles

Program Committee Member (Applied Sciences & Engineering); CS DEI Ambassador (Undergraduate) ; CS Student Association Officer (Graduate)

Community

Graduate Women in CS; Afro-Latinx (Vice President); CS+SG

Professional Affiliations

SACNAS; CVF; LatinX in AI

Conference Reviewer

WACV (2025); CVPR (2025)